

Variation in human intestinal microbiota with age

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The large intestinal microbiota plays an important role in normal bowel function and the maintenance of host health, through the formation of short chain fatty acids, modulation of immune system reactivity and development of colonisation resistance. However, the effects of ageing on bacterial community structure in the colon are not well documented. Aim of this study is to assess bacterial species diversity in the human faecal microbiota with respect to age and Clostridium difficile infection. Bacterial populations were quantified from stool samples obtained from children (16 months to seven years), young adults (21-34 years), healthy elderly people (67-88 years) and patients diagnosed with Clostridium difficile-associated diarrhoea (68-73 years). Microbial diversity was assessed to species level for samples from the latter three subject groups. Marked inter-individual variations occurred in microbial composition at genus and species levels. The faecal microbiota of children was found to be bacteriologically less complex whilst advancing age was associated with decreased bifidobacteria and increased bacteroides species diversity. Changes in microbial composition with age or disease will alter the metabolic capacity of the gut microbiota and has important implications for therapies aimed at modulating the large intestinal microbiota.

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Introduction

Factors such as diet and antibiotic therapy are of undoubted importance in modulating the composition and metabolic activities of the colonic microflora. However, the greatest changes in species composition occur during the natural process of bacterial succession in infancy. Ingested environmental microbes inoculate the previously sterile colon of newborn infants and the acquisition of an intestinal microflora begins, eventually developing to resemble that of adults.

Diet is thought to be a dominant factor regulating the sequence of colonisation, since the microflora of breast-fed infants can often differ from that found in bottle-fed babies¹. A diet of breast milk tends to favour the development of a simple microflora, consisting predominately of bifidobacteria, and while a diet of formula milk may also allow high bifidobacterial counts, bacteroides and clostridia colonise the bowel more frequently and facultative anaerobes attain higher cell densities^{2,3}. However, variation in the faecal flora of infants tends to be large, and bacteroides have been reported to be the dominant anaerobic bacteria irrespective of diet, with bifidobacteria being absent from more than half of the faecal specimens analysed^{4,5}. Such differences probably relate to the effects of inter-individual and environmental factors, and possibly the cultural techniques used to monitor bacterial populations.

Interestingly, *Clostridium difficile* (*C. difficile*) can be isolated from the fac-

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ces of neonates, and even though high levels of toxins A and B might be present, no symptoms of disease are displayed. At this age, the normal gut flora is underdeveloped and, therefore, unlikely to afford any protection in excluding this pathogen. One hypothesis for the asymptomatic carriage is that microbial degradation of host glycoproteins is required to expose toxin receptors in the gut, and thus susceptibility to *C. difficile* toxins parallels the development of the gut flora⁶. The age at which this neonatal protective mechanism is lost and colonisation resistance takes over as the protective mechanism against *C. difficile* infection remains unknown. However, the evolving microbial community within the large intestine is clearly of major consequence to the host.

The colonic microflora of infants can be described as 'adult-like' after the age of two years, although populations of facultative anaerobes are often observed to be greater than those of healthy adults^{2,7}. Indeed, it is likely that the intestinal microflora does not completely resemble that of adults until much later in childhood. Once the climax microflora has become established, the major bacterial groups in the faeces of adults remain relatively constant over time⁸. It has been noted, however, that elderly people harbour fewer bifidobacteria and higher populations of fungi and enterobacteria compared to younger adults⁹. Therefore, bacterial succession within the large bowel probably continues throughout life. Benno et al.¹⁰ found that the intestinal microflora of young dogs contained higher populations of bacteroides, bifidobacteria, lactobacilli, and anaerobic cocci, whilst older animals harboured greater numbers of clostridia and streptococci. These animals were all reared under similar conditions to minimise the influence of diet and environment, although the factors regulating these changes were not identified.

The frequency of isolation of *C. difficile* is also greater in the elderly. Whilst this is partly a result of factors such as hospitalisation, bacterial populations of the genus *Clostridium* have been found to be significantly higher in the faeces of healthy elderly volunteers compared to the young¹¹.

A number of physiological changes occur in the body with advancing age. These include decreased acid secretion by the gastric mucosa and greater permeability of mucosal membranes in the gut which have been linked to increases in circulating antibodies to components of the intestinal flora in elderly subjects¹². Thus, it can be assumed that certain bacterial strains take advantage of new ecological niches, thereby inducing a shift in the composition of the gut microflora.

Comparative sequence analysis of 16S rRNA indicates that a large proportion of the organisms present in the

large bowel have yet to be described. The advent of molecular methods which do not rely on our ability to culture bacteria prior to quantification allows additional information to be gained on the gut microflora as a whole. Dore et al.¹³ used such techniques to investigate bacterial succession in infants and found that culturable methods lead to an overestimation of the proportions of bacteroides and bifidobacteria. This method has also been applied to adult faeces with additional probe sequences¹⁴.

Another method that allows ecological analysis without the need to culture organisms is that of community cellular fatty acid (CFA) analysis. Numerous environmental factors affect bacterial fatty acid synthesis but certain signature fatty acids have been used to indicate the presence of specific groups of organisms in soil and marine environments, and have also been used to study community structure in human faecal samples.

Clearly, the microbial composition of faeces can vary greatly from the time of birth to old age. Here we compare the bacterial composition of faeces obtained from children, young adults, and elderly subjects using different analytical techniques. Also, given the importance of the large intestinal microflora in colonisation resistance against *C. difficile* infection, predominant bacterial populations in a group of elderly patients with *C. difficile*-associated diarrhoea were analysed. The bacteria were determined by viable count, 16S rRNA, and community CFA methodologies.

Subjects and Methods

Subjects

Faecal samples were obtained from a total of 22 subjects who had no history of gastrointestinal upset or antibiotic therapy within the previous two months. Ten were from children aged 16 months - seven years, seven from young adults aged 21-34 years and five from elderly people aged 67-88 years. An additional subset of samples was obtained from four elderly patients aged 68-73 years on the day they were diagnosed with *C. difficile*-associated diarrhoea (CDAD). All stool specimens were processed within 60 mins for viable bacterial counts, and samples were frozen for community CFA and 16S rRNA analysis at -20°C and -80°C respectively.

Bacteriological measurements

Faecal samples were homogenised, sieved and serially diluted before plating onto various selective agars for enumeration as previously described¹⁵. The agars used were as follows; Beerens agar¹⁶ (bifidobacteria),

Blood Azide agar (enterococci), Cycloserine-cefoxitin-fructose agar (*C. difficile*), MacConkey agar No. 2 (enterobacteria), Nutrient agar (total facultative anaerobes) Perfringens OPSP agar (*Clostridium perfringens*), Rogosa agar (lactobacilli) Wilkins-Chalgren agar (total anaerobes), Wilkins-Chalgren agar plus GN selective supplements (Bacteroides, Porphyromonas, Prevotella).

Bacterial cellular fatty acid analysis

Bacterial cellular fatty acids were identified using the Microbial Identification System MIDI (Newark, NJ, USA). The fatty acid methyl esters of strict anaerobes were extracted from bacterial cell mass obtained from approximately 30 ml of overnight culture in Peptone Yeast Extract broth while facultative anaerobes were cultivated using Trypticase Soy Broth Agar (TSBA) ¹⁷. For community fatty acid structure, CFA were extracted from bacterial cell mass in 1 ml of fresh faecal slurry. Methylated fatty acids were identified using gas chromatography as reported elsewhere ¹⁸ and a bacterial identification obtained by CFA profile comparison using standard libraries, Moore and TSBA for strict and facultative anaerobes, respectively.

16S rRNA analysis

Total rRNA was extracted from 0.5 ml aliquots of faecal slurries using the mechanical disruption and phenol:chloroform method described earlier ¹⁹. RNA was immobilised on nylon membranes and hybridised with radiolabelled oligonucleotide probes Bact338 ²⁰, Enterol432 ²¹, Bacto1080 and Bif1278 ¹³ as described previously ²². Specific wash temperatures were 54, 44, 46 and 46°C, respectively.

Statistical analyses

A one-way analysis of variance was employed to compare differences between each group of subjects, using Data Desk version 6 software (Data Description Inc., Ithaca, NY, USA).

Results

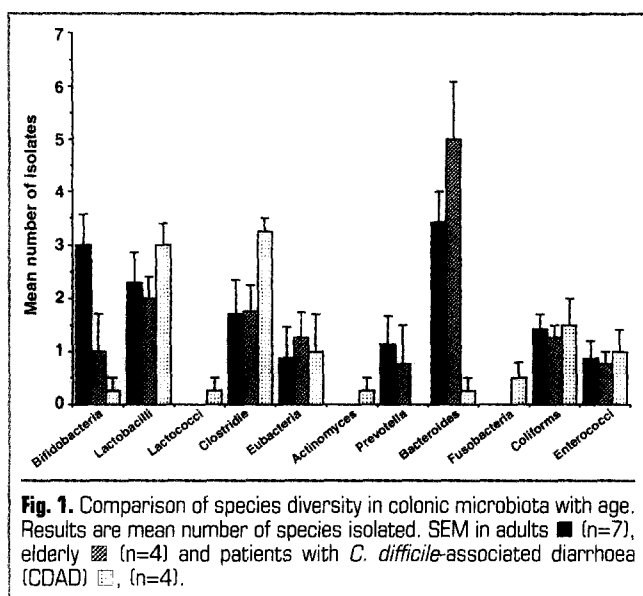
Bacteriology of different subject groups

Results in Table I show that whilst total anaerobe counts were similar in all four subject groups, bacterial compositions at genus level varied markedly. Populations of bacteroides-group organisms were significantly lower in CDAD patients compared to children, adults, and healthy geriatrics, whilst numbers of bifidobacteria were reduced in geriatric patients, irrespective of *C. difficile* infection. The species diversity of these organisms was notably less in the CDAD group, with only one species of bacteroides and one of bifidobacteria detected in this group of four patients (Fig. 1). The most prevalent bifidobacterial species were *B. angulatum* and *B. adolescentis* which were detected in the faeces of nearly all subjects with detectable bifidobacteria, whilst the range of bacteroides species was more diverse. In contrast to bacteroides and bifidobacteria, species variations in clostridia and lactobacilli did not decline in the CDAD group, with the total counts of clostridia and lactobacilli being significantly higher in the CDAD group compared to adults, and geriatrics respectively (Table I). Enterobacterial counts were lowest in the adult group and these populations increased markedly in children and CDAD patients. However, Figure 1 shows there was little change in species diversity in facultative anaerobic populations.

Table I. Total viable counts in faeces of different subject groups.

Bacterial group	Log ₁₀ cfu/g wet weight faeces			
	Children	Adults	Elderly	CDAD
Total anaerobes	10.4±0.2	10.5±0.1	10.1±0.2	10.0±0.3
Bacteroides group	9.9±0.4*	10.0±0.1†	9.6±0.2†	6.6±0.8*†‡
Bifidobacteria	9.8±0.3*†	9.1±0.2‡§	7.3±1.0*†	6.3±0.5‡§
Enterobacteria	8.0±0.4*	5.9±0.5*	6.7±0.8	8.4±0.2
Clostridia	7.2±0.8	6.6±0.4*	6.9±0.6	8.7±0.6*
Lactobacilli	6.6±0.7	6.7±0.6	5.4±1.0*	8.8±0.7*
Enterococci	5.5±0.5	6.1±0.7	6.0±0.8	7.3±1.2

Results are expressed as mean (SEM) in children (n=5), adults (n=7), elderly subjects (n=4) and patients with *C. difficile* associated diarrhoea (CDAD, n=4). Corresponding symbols in the same row denote means are significantly different (p≤0.05).



16S rRNA

Table II shows total bacterial rRNA peaked in the adult subjects, whilst levels in CDAD patients were significantly lower than in all other groups. Despite large variation, the bacteroides group probe revealed similar

levels of this type of rRNA in each subject group, with the exception of the CDAD patients in whom none was detected. Amounts of bifidobacterial rRNA were double those of bacteroides, and with the exception of the elderly group, followed a similar pattern, in that only small amounts were present in the faeces of CDAD patients. The faeces of elderly subjects were found to contain either negligible or high amounts of rRNA specific for bifidobacteria, and hence the standard error associated with this group was extremely large. Relatively low concentrations of enterobacterial rRNA were found in all groups, with the faeces of children containing twice that found in adults and geriatrics. However, the greatest levels of this type of rRNA were detected in CDAD patients.

Expressed as a percentage of total bacterial rRNA, enterobacterial rRNA was markedly higher in the CDAD group, both with a value approaching 10% of the total rRNA (Table III). The stools of children also contain significantly higher proportions of this type of rRNA when compared with the adult and elderly subject groups, although all were less than 2% of the total. The proportion of bacteroides group rRNA was still significantly lower in the CDAD patients. Despite retaining a similar pattern to results expressed as a concentration, variation in the percentage of bifidobacter-

Table II. Concentrations of 16s rRNA in faeces of different subject groups.

Specific rRNA probe	µg rRNA/g wet weight faeces			
	Children	Adults	Elderly	CDAD
Total bacteria	1621±344*	2327±486 [†]	1991±580 [†]	552±288* ^{††}
BPP group	248±58*	228±73 [†]	193±72 [†]	0±0* ^{††}
Bifidobacteria	616±205*	536±217 [†]	867±664	49±49* [†]
Enterobacteria	18±3* [†]	7±0* [†]	7±1 ^{†§}	29±8 ^{†§}

Results are expressed as mean (SEM) in children (n=10), adults (n=7), elderly subjects (n=5) and patients with *C. difficile* associated diarrhoea (CDAD, n=4). BPP, *Bacteroides*-*Porphyromonas*-*Prevotella*. Corresponding symbols in same row denote means are significantly different ($p \leq 0.05$).

Table III. Percentage of specific 16s rRNA in faeces of different subject groups.

Specific rRNA probe	% Total rRNA			
	Children	Adults	Elderly	CDAD
BPP group	14.8±1.7*	9.1±2.7 [†]	9.7±3.9 [†]	0.0±0.0* ^{††}
Bifidobacteria	39.6±10.6	22.1±7.0	29.1±18.8	19.9±18.2
Enterobacteria	1.3±0.2* ^{††}	0.3±0.1* [§]	0.7±0.3 ^{†§}	9.3±4.1 ^{†§}

Results are expressed as mean percent±SEM; children n=10, adults n=7, elderly n=5, CDAD n=4. CDAD, elderly patients with *C. difficile*-associated diarrhoea. Corresponding symbols in same row denote means are significantly different ($p \leq 0.05$).

ial rRNA within groups was seen to increase, particularly in the CDAD group which, as a consequence, was no longer significantly different from the other groups.

Cellular fatty acids

Community CFA analysis showed that whilst the straight chain saturated fatty acids 16:0 and 18:0 predominated, faecal profiles differed considerably between groups (Table IV). Patients with CDAD were characterised by high levels of 20:0, 20:1 cis11, low amounts of 12:0, 15:0 and absence of the branched chain 15:0 ante and 15:0 iso fatty acids. The highest levels of branched CFA (BCFA) were seen in the adult group, with the total averaging almost 5%. The faeces of adults were also found to contain a higher proportion of 18:1 cis11 dimethyl acid (DMA) and 14:0 DMA than those of children and CDAD patients and whilst only minor amounts of 14:0 iso were detected in the faeces of adults, its presence distinguished this group of subjects from the children and elderly subjects. Another DMA fatty acid, 15:0 ante DMA, was also a characteristic of adult stools. The faecal samples of children were found to be unique in their lack of two fatty acids, 18:0 DMA and 16:1 cis9.

Discussion

Viable counts of predominant bacterial species isolated from the faeces of adults and children showed some variation such as higher bifidobacterial and clostridial populations in the latter group. However, the only population to vary significantly was the enterobacteria which were 100-fold higher in the children's faeces. This is in agreement with previous studies showing that whilst the overall community structure is adult-like, facultative anaerobes often remain at elevated levels in the large intestine of infants². This finding was confirmed by rRNA analysis (Tables II and III). It is probable that as the microflora matures, competition from other bacterial species increases causing a reduction in facultative anaerobic populations. An interesting point to note is that rRNA analysis showed that the proportions of enterobacterial, bifidobacterial, and bacteroides-group organisms were elevated in the group of children thus showing that the intestinal flora of the children was bacteriologically less complex than that of the adults.

The number of bifidobacterial isolates was also found to decrease in healthy elderly, compared to young elderly, subjects at genus level, although this was not ob-

Table IV. Bacterial community CFA composition in the faeces of different age groups.

CFA	Children	Adults	Elderly	CDAD
Saturated straight chain				
12:0	1.0±0.4	4.8±4.0*	0.8±0.3 [†]	0.1±0.1 ^{*†}
15:0	1.7±0.5*	3.0±0.8 [†]	1.6±0.4 [†]	0.2±0.2 ^{*††}
16:0	43.9±2.9 ^{*†}	24.8±3.3 ^{*†}	37.0±3.8 [†]	30.3±5.6 [†]
18:0	24.1±2.3*	15.3±3.5*	21.9±1.7	16.9±2.7
20:0	2.9±1.1*	2.7±1.6 [†]	3.0±1.0 [†]	9.6±3.2 ^{*††}
Total	79.3±3.3 ^{*†}	58.3±7.5*	68.4±5.7	60.3±7.4 [†]
Unsaturated straight chain				
16:1 cis9	0.0±0.0 ^{*††}	1.1±0.4*	0.4±0.1 [†]	1.1±0.3 [†]
18:1 cis9	6.9±1.8*	8.2±2.3	6.6±2.0	15.5±4.2*
20:1 cis11	0.2±0.2*	0.2±0.2 [†]	0.2±0.2 [†]	2.9±1.3 ^{*††}
Total	14.6±2.5*	24.5±10.7	17.5±4.6	32.4±6.4*
Branched chain				
14:0 iso	0.0±0.0*	0.2±0.1 ^{*†}	0.0±0.0	0.0±0.0 [†]
15:0 anteiso	2.2±1.0*	2.1±0.5 [†]	1.1±0.3 [†]	0.0±0.0 ^{*††}
15:0 iso	0.4±0.2 ^{*††}	0.9±0.2 ^{*§}	0.7±0.2 ^{*††}	0.0±0.0 ^{†§†}
Total	3.3±1.2*	4.8±1.1 [†]	2.5±0.7 [†]	0.1±0.4 ^{*††}
Dimethyl acyl				
14:0 DMA	0.0±0.0 ^{*†}	0.6±0.2 ^{*†}	0.3±0.1 [†]	0.1±0.1 [†]
15:0 anteiso DMA	0.0±0.0*	0.5±0.1 ^{*††}	0.0±0.0 [†]	0.0±0.0 [†]
18:0 DMA	0.0±0.0 ^{*††}	1.1±0.3*	0.8±0.3 [†]	1.7±0.6 [†]
18:1 cis11 DMA	0.0±0.0 ^{*†}	1.1±0.3 ^{*†}	0.5±0.1 ^{†§}	0.1±0.1 ^{†§}
Total	2.3±0.5	7.5±1.9	4.8±0.6	3.8±0.3

Results are expressed as mean %±SEM; children n=10, adults n=7, elderly n=5, CDAD n=4. CDAD, elderly patients with *C. difficile*-associated diarrhoea. Corresponding symbols in same row denote means are significantly different ($p \leq 0.05$).

vious from the 16S rRNA data due to the high degree of variation between subjects observed using these procedures. *B. angulatum* was the most common bifidobacterial isolate in healthy young adults, but species diversity was found to decrease with *B. bifidum*, *B. catenulatum*, *B. pseudocatenulatum* and *B. infantis* not being detected in the faeces of elderly subjects (results not shown). Previous work has identified *B. adolescentis* as the most common bifidobacterial isolate in adults and centenarians²³ although *B. angulatum* was not listed and may have been overlooked due to the method of identification. The application of polymerase chain reaction (PCR) found the *B. catenulatum* group and *B. longum* to be more common than *B. adolescentis* in adults, although again *B. angulatum* was present in few subjects²⁴. This study showed that the adult microflora generally harboured a combination of 3 to 4 different bifidobacterial species, and our results are in agreement with this (Fig. 1). This could have important consequences for the application of prebiotics as prophylactics against intestinal infection or overgrowth.

Greater proportions of bifidobacteria were detected than in previous rRNA studies although this could, in part, reflect inherent differences in experimental methodology. Fluorescent in situ hybridisations have shown an overestimation of bifidobacteria compared to plate counts, and these organisms can be difficult to detect using dot blot analyses^{25,26}. Even so, the carefully standardised procedure used to extract rRNA from faeces allowed expression of the abundance of a specific microbial population as total quantities, facilitating a more confident expression of the contribution made by a bacterial population to overall community composition. Changes in population abundance, measured and expressed as a proportion of total rRNA may represent shifts in total community structure and ribosomal abundance, and not changes in absolute amounts.

The microbiota of CDAD patients was markedly different from all other samples with greatly reduced species diversity of bifidobacteria, prevotella and bacteroides with a concomitant increase in clostridia and lactobacilli. It was probably this reduction in some of the competing anaerobes, which would usually overwhelm agar plates, that limited the detection of the fusobacteria and actinomyces to CDAD patients rather than these organisms not being present in other subject groups. This is an inherent limitation of the viable count method and their enumeration would require the use of methods that circumvent the necessity to culture and identify organisms, such as FISH analyses.

The increase in facultative anaerobes in these subjects could be related to the effects of metronidazole therapy although viable count data revealed that high populations of anaerobic genera such as clostridia and lactobacilli were present at the time of sampling. Indeed,

enterobacteria could account for over 9% of the total metabolic potential in these faeces showing the severe perturbations in the large intestinal populations of this subject group.

The absence of bifidobacterial populations and very low levels of bifidobacterial rRNA in the CDAD group and some elderly subjects suggests prebiotic therapy could be beneficial in advanced age groups. However, given the low bifidobacterial species diversity observed in these subjects, simultaneous probiotic administration might be deemed advantageous in providing an initial increase in the total bifidobacterial counts whilst prebiotics aid recovery of indigenous populations. Also, this further highlights the need to identify the species-specificity exhibited by various prebiotic preparations in the large intestine.

In this study, we also monitored faecal community CFA to identify altered bacterial composition. The marked alterations in bacterial populations detected in the CDAD patients were revealed by a greatly altered CFA profile obtained from these stools. Since fatty acid synthesis is strongly influenced by growth rate and environmental conditions^{22,27}, these changes were probably due, in part, to the increased transit time that accompanies *C. difficile* infection. However, it is likely that variation of certain fatty acids such as 20:1 cis11 which is a component of some clostridia could be attributed to changes in bacterial population size. One good example, signature fatty acids would be BCFA, particularly 15:0 ante, levels of which correlated well rRNA and viable counts for bacteroides.

DMA analogues were lower in the CDAD group compared to those in adult and elderly subjects, and the reduction in 18:1 cis 9 DMA and 18:1 cis11 DMA could be taken as evidence of reduced bifidobacterial populations¹⁸. However, the association of DMA formation with low specific growth rates²² indicates these reductions could also be attributed to increased transit time in the colon. Conversely, the largest proportion of 18:0 DMA was detected in the faeces of CDAD subjects. Whilst this increase was not significantly higher than the adult and elderly groups, the association of this CFA with other bacterial species such as clostridia could be implicated as a reason for this increase and, therefore, would not be a reliable marker of bifidobacteria in faecal samples.

Hence, the use of signature CFA to monitor bacterial populations in faecal samples was found to be effective for BCFA and bacteroides, whilst the link between DMA moieties and bifidobacteria, and 20:1 cis11 and clostridia, retains a certain degree of ambiguity. Ribosomal RNA proved a useful tool in the analysis of the human faecal flora, and although this method was not as rapid as the CFA technique, the specificity was much greater and, therefore, deemed superior.

In summary, the gut microflora is not completely adult-like until much later than previously assumed and probably continues to change throughout the life of an individual. The altered composition of metabolically active groups such as bacteroides and eubacteria probably leads to profound changes in the biochemical capacity of the gut microbiota with advancing age, and such changes might not be obvious from bacteriological studies made at genus level. In particular, the altered species diversity could have important implications for prebiotic therapy where the choice of carbohydrate would need to target a species that is likely to be present in members of the subject group. The reduced number of bifidobacterial isolates present in elderly subjects suggests a combination of pre- and probiotic treatment may be more likely to succeed as therapy for *C. difficile* infection in the elderly. The advent of new methodologies is leading to great increases in our knowledge of the large intestinal microbiota. Each technique has advantages and inherent limitations, such as limited nucleic acid probe diversity or low signature fatty acid specificity, but each is capable of contributing to our understanding of this complex ecosystem provided we take a systemic approach and weight each method accordingly.

List of abbreviations

BCFA: branched chain fatty acid; BPP: *Bacteroides porphyromonas* prevotella; CDAD: *Clostridium difficile*-associated diarrhoea; CFA: cellular fatty acid; DMA: dimethyl acyl; PCR: polymerase chain reaction; TSBA: Trypticase Soy Broth Agar

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